Digital Image Fundamentals & Sampling

Visual perception, light, electromagnetic spectrum, sampling and quantization.

1. Elements of Visual Perception & Image Formation

- Structure of human eye: cones (photopic vision) and rods (scotopic vision).
- Light spectrum: visible wavelength range 400nm (violet) to 700nm (red).
- Image acquisition pipeline: sensors, illumination source, and digital conversion.

2. Image Sampling and Quantization

- Spatial resolution: number of pixels per unit length (sampling density).
- Intensity resolution: number of discrete intensity levels (2^k bits per pixel).
- Aliasing and Nyquist-Shannon sampling theorem in 2D spatial frequencies.